

Harshal Dev

✉ harshaldev6@gmail.com ☎ 9818800898  [LinkedIn](#)

Education

Indraprastha Institute of Information Technology (IIIT) Delhi
B.Tech in Computer Science Engineering

Aug 2019 – May 2023
CGPA: 8.43/10

Technical Skills

Programming: C++, Java, Python, JavaScript, Golang
Frameworks & Databases: Spring Boot, Spring Webflux, Spring Cloud, PostgreSQL, MongoDB, MySQL, Redis
Backend & System: Distributed Systems, Microservices, REST APIs, Websockets, Apache Kafka, gRPC, Protobuf, Netflix Eureka
Concepts: Data Structures & Algorithms, OOPs, Operating System, DBMS, Design Patterns

Experience

Software Engineer - Jio Platforms Limited
Bengaluru, India

June 2023 – Present

JioTalkie

- Designed and developed scalable **microservices** using **Java, Spring Boot**, supporting **10,000+ users** and **reducing** average **API response latency by 18%**.
- Developed a **Kafka-based analytics pipeline** to capture and process **100K+ daily events** from distributed microservices, enabling reliable event processing, and near real-time dashboard reporting.
- Built a **notification service** supporting **asynchronous** multi-channel delivery (Push, SMS, Email) with retry, idempotency, and fallback mechanisms for reliable downstream communication.
- Contributed to **high-availability (HA)** architecture for distributed microservices using **Netflix Eureka** clustering, Spring Cloud Gateway, load balancing and rate limiting, improving service resilience during downstream failures.
- Improved **PostgreSQL** performance through query plan optimization, indexing, normalization, partitioning, and efficient schema design, reducing analytics query execution time by **40%**.
- Designed and maintained secure **RESTful APIs** with versioning, validation and documentation, while integrating **HashiCorp Vault** for centralized secrets management and secure credential handling across distributed systems.

JioPay

- Worked on JioPay's **voice confirmation feature**, implementing backend logic to process payment status updates using **MQTT** and enable real-time event-driven transaction handling.
- Designed and developed **REST APIs** and validation mechanisms to trigger voice confirmations for successful and legitimate payment events, ensuring correctness and preventing duplicate alerts.

Projects

PubSub | [GitHub](#)

Jan 2023 – Apr 2023

- Developed **distributed publish-subscribe system** allowing clients to publish & fetch articles from connected servers.
- Utilized **Protobuf** for service/message definition and **gRPC** for efficient server-client communication.
- Implemented **DFS-based recursive article fetching** and **cycle detection** for new server addition.
- Tools and Technologies:** Python, gRPC, Protobuf

Accomplishments

- Selected for **multiple leadership and mentorship roles**, including Teaching Assistant for Probability & Statistics, Student Mentor, and Hostel Committee member.